

ML-DL-Ops Assignment-1

Report

-By Shivam Khanchandani (B22BB038)

Colab links:

Q1A:

https://colab.research.google.com/drive/1dhZkGfo326Bgl3D_FkK_sd7rlw47hODH#scrollTo=hHwc3D0eEyPm

Q1B:

https://colab.research.google.com/drive/1xM7eG7sXy5UuZ4lZRAgPWz_okq3mg6B3#scrollTo=1Qj8ldBmrMQN

Q2:

https://colab.research.google.com/drive/18ryMkaH9drI6y8qLEntI_9Veg7WlhdpA#scrollTo=gh51rptWTb_F

Best Model:

<https://colab.research.google.com/drive/1n4Y4Y6juq1tacxECjZqctC2pFMXDEHJW#scrollTo=ZXr82uvs7EPF>

Github link:

<https://github.com/SK-038/MLOps-ShivamKhanchandani-B22BB038/tree/assignment-1>

Q1a) ResNet-18 and ResNet-50 models on MNIST and FashionMNIST datasets

Results:

Dataset: MNIST								
Batch Size	Optimizer	Learning Rate	ResNet-18 Accuracy (%)	No. of Epochs	pin memory	ResNet-50 Accuracy (%)	No. of Epochs	pin memory
16	SGD	0.001	97.44	5	True	97.36	5	True
16	SGD	0.0001	77.62	5	False	53.98	5	False
16	Adam	0.001	99.16	3	False	98.91	5	True
16	Adam	0.0001	99.3	5	True	99.15	5	True
32	SGD	0.001	96.21	5	True	95.07	5	True
32	SGD	0.0001	63.64	5	True	40.69	5	True
32	Adam	0.001	99.0	5	False	98.77	5	False
32	Adam	0.0001	99.14	5	True	98.94	3	True

Dataset: FashionMNIST							
Batch Size	Optimizer	Learning Rate	ResNet-18 Accuracy (%)	No. of Epochs	pin memory	ResNet-50 Accuracy (%)	No. of Epochs
16	SGD	0.001	88.01	5	False	87.56	5
16	SGD	0.0001	74.56	5	True	74.43	5
16	Adam	0.001	93.46	5	False	90.05	5
16	Adam	0.0001	93.26	5	True	91.59	5
32	SGD	0.001	80.36	5	True	79.87	5
32	SGD	0.0001	69.39	5	False	67.32	5
32	Adam	0.001	92.57	5	False	91.67	5
32	Adam	0.0001	92.62	3	False	90.98	5

I performed experiments using:

- Resnet18, Resnet50 (Models)
- Adam, SGD (Optimizers)
- MNIST, FashionMNIST (Datasets)
- 16, 32 (Batch sizes)
- 0.001, 0.0001 (Learning rates)
- 3, 5 (No. of epochs)

- True, False (pin memory)

This question explores the relationship between model depth, optimizer choice, dataset complexity and hyperparameters like batch size, learning rate, no. of epochs, pin memory.

Observations:

- **Dataset Difficulty Gap:** Performance on MNIST is consistently higher as compared to FashionMNIST because FashionMNIST contains more intricate textures and overlapping shapes.
- **Optimal Depth for Simple Tasks:** ResNet-18 often achieves higher accuracy than ResNet-50 on these datasets. Deeper models like ResNet-50 can overfit or struggle with convergence on low-resolution grayscale images, whereas ResNet-18 has a better depth-to-complexity ratio for this specific task.
- **Optimizer Dominance:** Adam provides superior performance, reaching high stability at a learning rate of 0.001. Conversely, SGD at low learning rates (0.0001) shows a significant accuracy drop, indicating it is highly sensitive to hyperparameter tuning and often stalls early in training.
- **Batch Size Sensitivity:** Smaller batch sizes (16) generally lead to better generalization in these tests compared to larger batches (32), as the inherent noise in smaller batches helps the optimizer escape local minima.

Q1b) SVM Model: RBF vs. Poly Kernels

Results:

Q1)B) SVM Classifier Results			
Requirement: Testing Classification Accuracy and training time in ms.			
Dataset	Kernel	Test Accuracy (%)	Train Time (ms)
MNIST	poly	97.71	288743.64
MNIST	rbf	97.92	268346.94
FashionMNIST	poly	86.30	473309.68
FashionMNIST	rbf	88.29	389151.25

I performed experiments on:

- Radial Basis Function (RBF), Polynomial (Kernels)
- MNIST, FashionMNIST (Datasets)

This question evaluates classical machine learning performance on the same vision tasks as before, by comparing kernel performance using Support Vector Machines (SVM).

Observations:

- **Kernel Superiority:** The RBF (Radial Basis Function) kernel consistently outperforms the Polynomial kernel in both accuracy and training efficiency. This is because RBF can map data into a higher-dimensional space more flexibly to find non-linear decision boundaries.
- **Dataset Complexity:** Accuracy on FashionMNIST is significantly lower for both kernels compared to MNIST, copying the trend seen in deep learning models.
- **Training Time Trade-offs:** While SVMs are faster than deep ResNets for smaller datasets, their training time scales poorly with dataset size, making them less scalable for large-scale vision tasks compared to CNNs.

Q2) FLOPs and Hardware Performance

Results:

Q2) Hardware Performance Analysis (FashionMNIST)							
Requirement: Comparison of CPU vs GPU performance, including FLOPs. All were run for 5 epochs							
Compute	Batch Size	Optimizer	Learning Rate	Model	Test Accuracy (%)	Train Time (ms)	FLOPs (GFLOPs)
CPU	16	SGD	0.001	ResNet-18	84.69	506547	0.0332
CPU	16	SGD	0.001	ResNet-50	90.73	1898165	0.0788
GPU	16	SGD	0.001	ResNet-18	85.08	446587	0.0332
GPU	16	SGD	0.001	ResNet-50	90.65	1907845	0.0788
CPU	16	Adam	0.001	ResNet-18	92.16	456461	0.0332
CPU	16	Adam	0.001	ResNet-50	93.54	2508762	0.0788
GPU	16	Adam	0.001	ResNet-18	92.07	408916	0.0332
GPU	16	Adam	0.001	ResNet-50	93.42	2376544	0.0788

This question discusses the computational "cost" of the models as seen in the above Hardware Analysis table.

Observations:

- **Fixed Theoretical Workload:** GFLOPs are a fixed property of the model architecture. ResNet-18 consistently requires 0.0332 GFLOPs, while ResNet-50 requires 0.0788 GFLOPs, regardless of whether they are run on a CPU or GPU or which optimizer is used.
- **Hardware Acceleration (Speed vs. Accuracy):** Training on a GPU is significantly faster compared to a CPU. Crucially, the Test Accuracy remains nearly identical across hardware types, proving that compute device only affects speed, not model capability.
- **Diminishing Returns on Compute:** ResNet-50 requires more than double the GFLOPs and significantly more training time than ResNet-18, yet on these datasets, it yields lower accuracy. This highlights that "bigger is not always better" for simple image classification tasks.

Best Model Graphs and Performance for both datasets

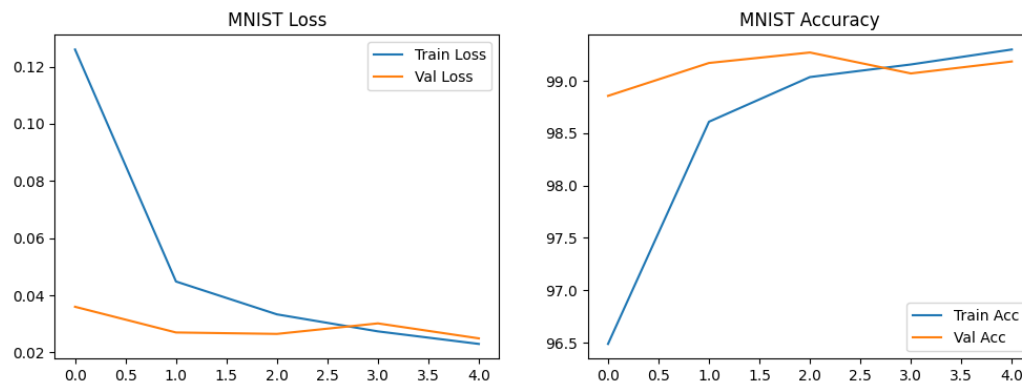
1) As can be inferred from the above tables, the best model for MNIST dataset was as follows:

- Model: Resnet-18
- Batch size: 16
- Optimizer: Adam
- Learning rate: 0.0001
- No. of epochs: 5
- Pin memory: True

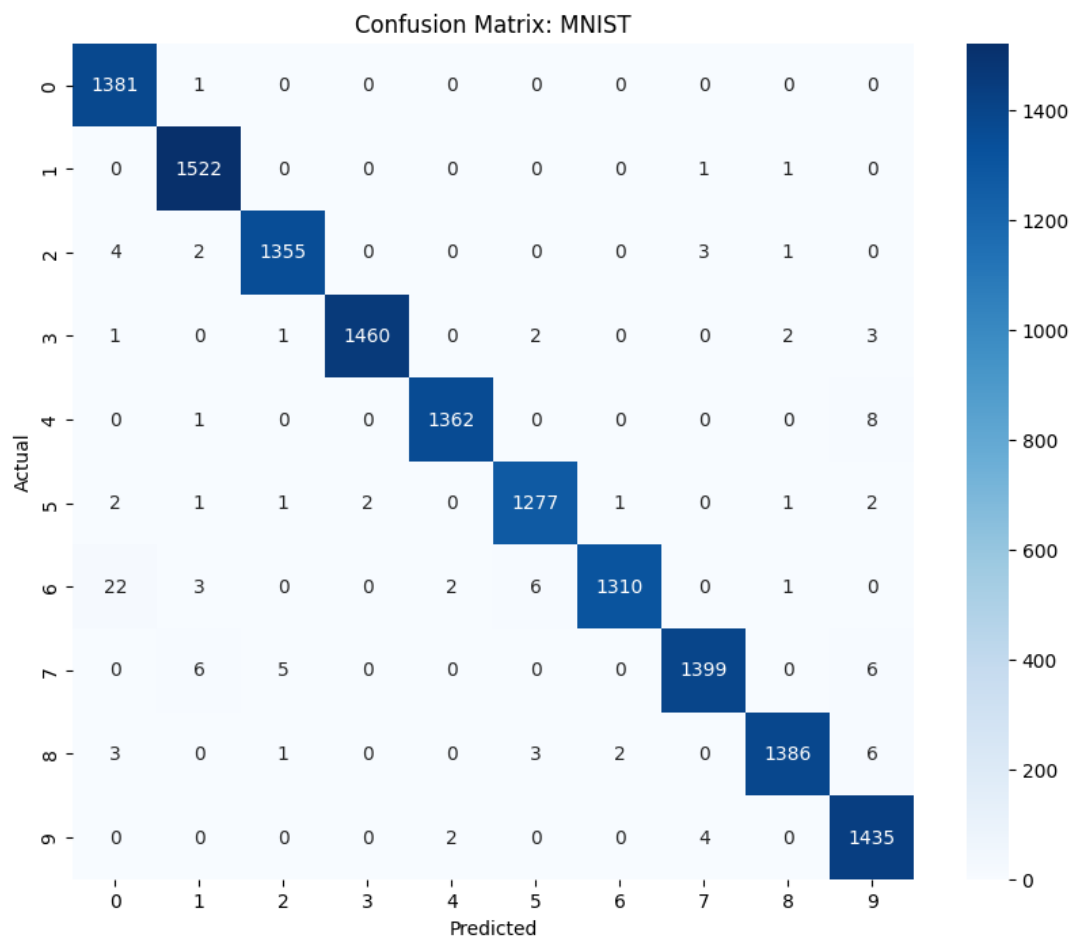
Test accuracy recorded: 99.3%

I ran this model with the given hyperparameters on the MNIST dataset separately to get the following graphs and results:

Training-Validation Graph:



Confusion Matrix:



2) As can be inferred from the above tables, the best model for FashionMNIST dataset was as follows:

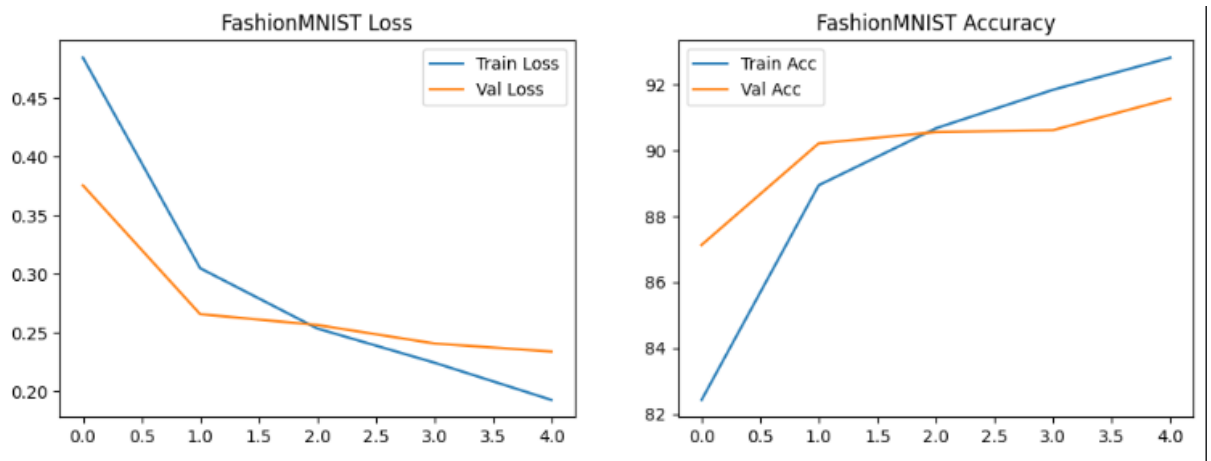
- **Model: Resnet-18**
- **Batch size: 16**
- **Optimizer: Adam**
- **Learning rate: 0.001**

- No. of epochs: 5
- Pin memory: False

Test accuracy recorded: 93.45%

I ran this model with the given hyperparameters on the FashionMNIST dataset separately to get the following graphs and results:

Training-Validation Graph:



Confusion Matrix:

